Whether Small Transformers Copy a Feature from Context Depends on the Noun's Training History

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Abstract

A language model can predict a word's properties by copying them from context or recalling what it stored. Across populations of items, bursty items whose meaning changes favour copying; we test this word by word within one model. We trained 230 small transformers on an artificial language, controlling how often new nouns occurred, with which shapes, and whether their sentences came singly or in bursts. Where object kinds keep their shapes, nouns lost their copying of a shape from the preceding sentence with exposure, unless their own earlier sentences had predicted a shape that memory could not. In a pre-registered replication, nouns in bursts whose shape varied between bursts copied 3.7 nats more after 1,000 sentences than nouns in fixed-shape bursts (4.2 nats at 85.6M parameters). The effect of a storable shape appears to arise in training: at test, a stored shape lowered copying only slightly (0.25 nats), though the model preferred the stored shape. What bursts kept was mostly a tendency to copy the latest shape rather than a link between noun and shape, and it spread to other nouns. Never-seen nouns copied mainly because their embeddings were untrained. In Pythia, replacing embeddings had no consistent effect.

1 Introduction

When a language model reads *The car was red. The car was*, it can predict *red* because the context says so, or predict a typical car colour from what it has stored. Which source it relies on matters for how models handle documents that contradict their knowledge (Longpre et al., 2021; Xie et al., 2024), for how they learn new words and tokens, and for how we interpret probes that use invented words. Work on in-context and in-weights learning shows that the balance depends on the training data. Classes that occur in bursts, or whose meaning changes, favour learning from context; common, stable classes favour learning stored in the weights (Chan et al., 2022a; Chan et al., 2025). Inside pretrained models, mechanisms that copy from the context compete with mechanisms that recall stored facts (Yu et al., 2023; Jin et al., 2024; Ortu et al., 2024). These results describe populations of items. We ask the same question about a single word: *what in a word's own training history decides whether a model copies its features from the context?*

Three accounts are plausible, and they make different predictions.

- **Memory competition.** When the model predicts, a stored feature competes with the context. Words with a stored feature should copy less than words without one.
- **Representation.** New words copy because their embeddings are untrained. Once a word's embedding looks like those of familiar words, copying should stop, whatever the word was seen with.
- **Useful context.** During training, the model learns for each word whether copying has paid off. A word should keep being copied if its own earlier mentions predicted its features and memory could not.

Natural data cannot separate these, because frequent words are both familiar and well known. Invented (nonce) words, the usual probe (Thrush et al., 2020; Misra et al., 2023), are new in both respects at once, and conclusions drawn with them can depend on how the new word is represented (Kim et al., 2022; Cacioli, 2026).

We therefore built the separation into training. Small transformers learned an artificial language describing objects by shape, colour and texture, into which we inserted 40 new nouns. For each noun we set how often it occurred, which shapes it occurred with, and whether its sentences came one at a time or in short bursts (Figure 1). We measured how strongly each model copied a shape named in one sentence into the next mention of a

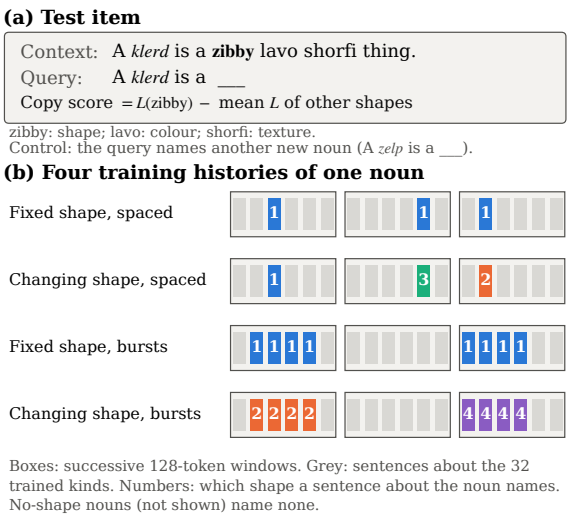


Figure 1: (a) A test item and the copy score, which leaves out the noun's own shape. (b) Four of the training histories, drawn schematically over three training windows; with the same number of sentences, bursts occupy a quarter as many windows as spaced sentences.

noun. Three pre-registered experiments, the third a replication on new models, separate the accounts.

The results join two of them. Unseen nouns copy because their embeddings are untrained. Familiar nouns lose copying unless their own earlier sentences were the only source of their current shape, as in bursts whose shape changes between bursts. Memory mattered, but its effect appears to arise in training: where a shape could be stored, the context was not needed, and copying was lost. At test, a stored shape lowered copying only slightly, although the model preferred the stored shape. What useful bursts kept was mostly a tendency, carried by the noun's embedding, to copy the latest shape rather than a link between noun and shape, and it spread to other nouns. Our claims are about small transformers trained on this language; we note parallels with children's word learning in Section 5.

2 Related work

In-context and in-weights learning. In-context learning emerges when items occur in bursts, more when their meanings change, and trades off against learning in the weights (Chan et al., 2022a); what is learned in context also generalises differently (Chan et al., 2022b). It depends on task diversity (Raventós et al., 2023), favours rare classes and can fade as stored solutions take over (Chan et al., 2025; Singh et al., 2023). Its onset has been traced to induction heads (Olsson et al., 2022; Reddy, 2024), and in small transformers stored bigram statistics are learned quickly and in-context ones later (Bietti et al., 2023). Synthetic languages and Markov chains have been used to trace how in-context learning forms (Akyürek et al., 2024; Edelman et al., 2024). Resetting embeddings during training keeps in-context use alive for rare tokens (Anand et al., 2025), and in synthetic biographies frequent entities shift from context to stored knowledge (Kim et al., 2026). We manipulate frequency, stored knowledge and arrangement separately for single words within one model.

Copying and recall. Language models have long gained from copying recent words, through caches and pointers (Kuhn and De Mori, 1990; Grave et al., 2017; Merity et al., 2017), because words recur in bursts (Church and Gale, 1995; Church, 2000; Madsen et al., 2005), a regularity that human memory also tracks (Anderson and Schooler, 1991). Pretrained models answer from memory more for frequent entities (Mallen et al., 2023; Kandpal et al., 2023; Razeghi et al., 2022) and resist contexts that contradict what they know (Longpre et al., 2021; Xie et al., 2024; Du et al., 2024); copying and recall compete inside the network (Yu et al., 2023; Jin et al., 2024; Ortu et al., 2024).

New words and new tokens. Rare and new words have poor representations (Schick and Schütze, 2020), and good embeddings for them can be built from a few contexts (Herbelot and Baroni, 2017; Khodak et al., 2018); under-trained tokens behave abnormally (Land and Bartolo, 2024), new embeddings are often initialised at the mean of existing ones (Hewitt, 2021), and tying input and output embeddings (Press and Wolf, 2017; Inan et al., 2017) trains every token's embedding as an output. Lexical exposure can inflate apparent generalisation (Kim et al., 2022), and when language models acquire a word depends mainly on its frequency (Chang and Bergen, 2022). Children's word learning offers related evidence on shape (Landau et al., 1988; Smith et al., 2002), on expectations about how features vary (Kemp et al., 2007; Dewar and Xu, 2010) and on the timing of exposures (Childers and Tomasello, 2002; Schwab and Lew-Williams, 2016), to which we return in Section 5.

3 Setup

Language. Each corpus describes 32 kinds of object, each named by an invented noun (the *trained*

History	Sentences about the noun in training
Unseen	none: the noun's embedding is untrained
Fixed shape	always the noun's own shape, which it can store
Changing shape	a new shape every sentence (spaced) or every burst (bursts), so nothing can be stored
No shape	no shape at all: the noun is familiar but has no shape to store

Table 1: Noun histories. In changing-shape bursts the shape is shared within a burst and redrawn for each burst, so the noun's own earlier sentence is the only source of its current shape.

Exp.	Nouns	Exposures	Models
1	fixed, changing, no shape; spaced	10–10,000	90
2	the same; spaced and bursts	100, 1,000	60
3	as Exp. 2 with new seeds; 85.6M models; single-history corpora	100, 1,000	70

Table 2: Experiments. Experiments 1 and 2 used stable-kind and shifting-kind corpora (Experiment 1 also random-order corpora); Experiment 3 used stable-kind corpora only. Single-history corpora: all familiar nouns have changing shapes, or all fixed shapes.

kinds), with 14 exemplars per kind. An exemplar has a shape, a colour and a texture, each one of ten invented words, and is described in one of five sentence frames, such as *A wug is a zibby lavo shorfi thing* (shape, colour, texture) or *Look at the wug. It is zibby and lavo* (Appendix A). The corpus runs through all exemplars many times, each pass arranged afresh, and a model reads it once. Sentences about a kind come in runs of four. In **stable-kind** corpora each kind keeps one shape, so its shape can be stored. In **shifting-kind** corpora a kind gets a new shape in every pass, so its current shape can come only from the context. Every word is one token.

Models. Decoder-only transformers with pre-layer normalisation and learned position embeddings, with 3.4M, 10.9M and 25.6M parameters (4 to 8 layers), and in Experiment 3 also 85.6M (12 layers), read the text once in windows of 128 tokens. Input and output embeddings are untied, so a noun's input embedding carries everything the model has learned about the noun as an input; we chose this after finding that earlier models in this language had been trained untied by a hardware fault (Appendix B). We also check tied embeddings.

Controlling a noun's history. We added 40 new nouns to each corpus. Four never occurred in training (*unseen*); the other 36 did (*familiar*). Each familiar noun had one history (Table 1) and a set number of sentences (*exposures*), spread evenly from 5% to 95% of training, with three nouns per condition. Presented *spaced*, a noun has one sentence per window. Presented in *bursts*, it has four consecutive sentences in one window, like the trained kinds' runs.

Three experiments (Table 2). Experiment 1 tested the memory and representation accounts with spaced nouns. In it, familiar nouns never recurred within a window, unlike the trained kinds, and no-shape nouns never appeared in the test frame. Experiment 2 removed both differences and added bursts, to test the useful-context account. It used 100 and 1,000 exposures, where Experiment 1's copying had nearly vanished. One side effect: no-shape nouns, which now appeared in the test frame without a shape, learned to expect a colour after *is a* (by 1 to 5 nats), so comparisons with them are confounded with word class. Experiment 3 was registered after Experiment 2's results to test them on new models: ten new seeds at the three sizes, ten models of 85.6M parameters, and five seeds in single-history corpora. New nouns fill about 60% of training windows in Experiment 1 and 8% in Experiments 2 and 3 (40% and 5% for the 10.9M models, which read more windows), so we compare conditions only within an experiment.

Measures. A test item gives one sentence about a noun naming a shape s, then the start of the next mention, *A noun is a*, and reads the model's log probability L of each of the ten shapes at the next position (Figure 1a). The **copy score** is $L(s)$ minus the mean of L over the other shapes, leaving out the noun's own shape s^*, which the context never names (nouns without a shape get a fixed stand-in). It is in nats: 3.4 means the context's shape is about 30 times as likely as the average other shape. Each noun gets 36 items. Because the copy score leaves out s^*, we also report which shape the model prefers, $L(s^*) - L(s)$, and how strongly it has stored s^*: $L(s^*)$ minus the mean of the other shapes after an unrelated sentence (**stored preference**). In a **control**, the query names another new noun with the same history; if copying is tied to the noun, the control should score lower. We also score unseen nouns with a **typical embedding**: the mean of the trained kinds' embeddings plus a random offset of typical length.

Comparison	Mem.	Rep.	Useful	Exp.
Unseen: typical − untrained		<0		–
Spaced: fixed − changing shape	<0	0	0	3
Changing shape: bursts − spaced	0	0^b	>0	2, 3
Bursts: changing − fixed shape	$>0^c$	0^b	>0	3

Table 3: Predictions of the memory-competition, representation and useful-context accounts for the copy score. Blank: no prediction. [b]Or >0 if an embedding left less trained copies more. [c]No larger than for spaced nouns. Exp.: experiments in which the comparison was a pre-registered test.

Predictions. Table 3 gives what each account predicts. Only memory competition predicts that a stored shape lowers copying among spaced nouns. Only useful context predicts that changing-shape bursts copy more than changing-shape spaced nouns, unless bursts help simply by updating the embedding less often. Fixed-shape bursts test that: they get as many updates, but their shape can be stored, so the context is not needed. Memory competition also predicts that they copy less, but only by as much as a stored shape lowers copying among spaced nouns.

Analysis. The seed is the unit: we average each condition over model sizes within a seed and test across seeds, with Holm correction within each registered family (Holm, 1979), and count a prediction as supported only if its direction also holds in two of the three sizes. Each experiment was registered before any of its models was trained. Where Experiment 3 tested a finding of Experiment 2, we report the Experiment 3 value; Table 4 gives both. Appendix C lists every registered test and its outcome.

4 Results

The confirmatory core is Experiment 3's registered replication (Sections 4.3–4.7 and 4.9): changing-shape bursts copy more than spaced and fixed-shape bursts, also at 85.6M parameters; a stored shape lowers copying slightly but wins in prediction; in mixed corpora the copying is mostly not tied to the noun, and the queried noun's history matters more than the context noun's; the effect exceeds what embedding geometry predicts; and it appears without nouns of other histories. The mechanism analyses, shifting-kind corpora, tied embeddings and natural text are exploratory. Results not marked pre-registered are exploratory.

4.1 New nouns copy because their embeddings are untrained

In stable-kind models of Experiment 1, unseen nouns had a copy score of 4.53 nats [95% CI 3.54, 5.52]. Given a typical embedding, the same nouns scored -0.33 (Figure 2, left); in Experiment 3 the drop was from 6.88 to 0.79. To see which property of an untrained embedding matters, we changed only the embeddings of unseen nouns in the Experiment 1 models. Scaling one to the trained kinds' average size, keeping its direction, lowered the score from 4.53 to 4.00 (pre-registered and supported, almost entirely in the smallest models). A random direction of that size kept copying (4.09), whereas the trained kinds' mean embedding removed it (-0.44). Size thus matters a little; what an embedding shares with trained nouns' embeddings matters most. With tied embeddings, which train unseen nouns' embeddings as outputs (ten 3.4M models), unseen nouns still copied (3.96 nats), a typical embedding still removed it (0.02), and changing-shape bursts still copied more than fixed-shape bursts at 1,000 exposures (1.90 [1.20, 2.60]), though all effects were smaller (Appendix F). In these small models, then, a probe with an unseen word measures the word's embedding as much as anything learned about context.

4.2 Nouns met one at a time lose copying

Changing-shape nouns never have a shape to store, so they are the cleanest test of memory competition. In stable-kind models of Experiment 1 they lost copying almost as quickly as fixed-shape nouns, which stored their shape well (stored preference 9.5 nats at 1,000 exposures; Figure 2, left). Fixed-shape nouns did not lose copying faster than no-shape nouns (pre-registered prediction, not supported). For all three histories, copying fell from 3.6–4.5 nats at 10 exposures to under 1 nat at 100 and to near zero from 1,000 on. No-shape nouns ended within the registered bound of unseen nouns with a typical embedding (0.33 nats [0.13, 0.52]; pre-registered). In random-order corpora no noun copied, so copying is learned from the runs of sentences about a kind. Once a noun was familiar, it lost copying largely whatever it had stored, as the representation account predicts.

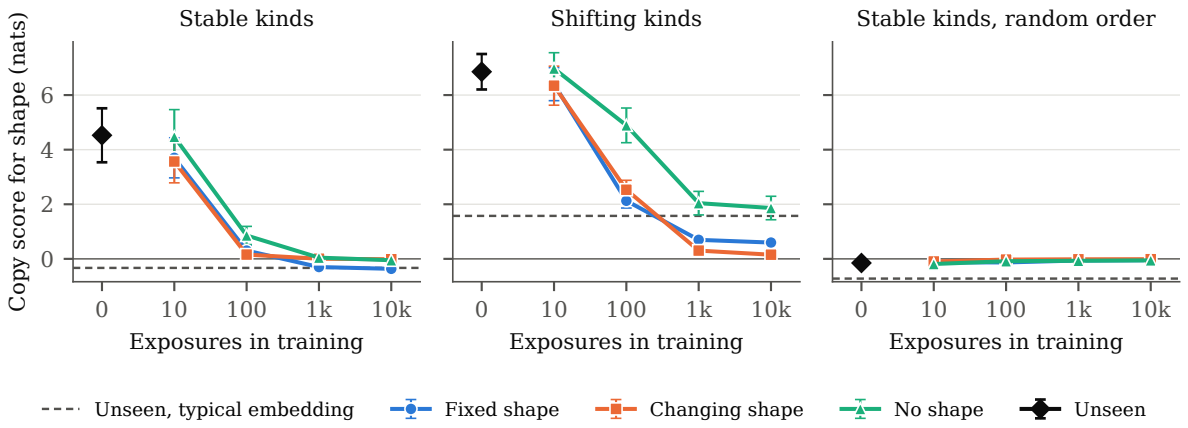


Figure 2: Experiment 1: copy score by number of exposures and noun history, all spaced, in stable-kind, shifting-kind and random-order corpora (means over ten seeds, sizes averaged; 95% intervals). Diamond at 0: unseen nouns. Dashed line: unseen nouns with a typical embedding.

4.3 Bursts with changing shapes keep copying

Changing-shape bursts, where a noun's previous sentence gives its current shape and memory cannot, scored 4.14 nats [3.60, 4.69] more than changing-shape spaced nouns (pre-registered; Figure 3a; Table 4), as much at 1,000 exposures as at 100. After 1,000 exposures they still scored 4.71 nats, against at most 1.1 for every other familiar history and 0.79 for unseen nouns with a typical embedding. Over training this copying did not fade: in Experiment 2, for nouns with 1,000 exposures, it rose from 0.8–2.6 nats at the first saved weights to 3.85 at the end, while every other history ended near or below 1 (Figure E4).

Fixed-shape bursts have the same frame, presentation and number of updates, but a shape that memory can supply. At 1,000 exposures changing-shape bursts copied 3.68 nats [2.99, 4.37] more than fixed-shape bursts, at all three sizes (pre-registered). Memory competition is unlikely to explain this. Among spaced nouns at 1,000 exposures a stored shape lowered copying by only 0.10, so the burst difference exceeded the spaced difference by 3.58 nats [2.94, 4.22]; even a proportional reduction of the same size (about a fifth) would predict a burst difference of about 0.9. Two further results point the same way. At 1,000 exposures no-shape bursts, which had nothing to store, copied no more than fixed-shape bursts (1.05 against 1.03), and in the tied-embedding models a stored shape did not lower spaced nouns' copying at all (−0.01). The results fit the view that what decided copying was whether the context had been needed in training, although our design did not manipulate memory during training directly.

	Exp. 2		Exp. 3	
Comparison	Pooled	1,000	Pooled	1,000
Changing: bursts − spaced	3.39^{r}	3.35	4.14^{r}	4.15
Bursts: changing − fixed	1.73	2.86	2.52	3.68^{r}
No shape: bursts − spaced	0.80^{r}	0.05	0.69	0.20
Spaced: fixed − changing	−0.02	0.02	−0.25^{r}	−0.10
Spaced: fixed − no shape^{a}	−1.02^{r}	−0.46	−1.00	−0.39
Same noun − other noun	0.05	0.05	0.07	0.08^{r}
Above geometry		2.84		3.67^{r}

Table 4: Key comparisons in stable-kind corpora (copy score, nats; seed-level means; 100 and 1,000 exposures pooled, and 1,000 alone). ^{r}Pre-registered test; all other values exploratory. Same noun − other noun: the control, for changing-shape bursts. Above geometry: changing-shape bursts relative to the copying predicted from their embeddings' geometry (per-model regression, seed-level). ^{a}Confounded with expected word class.

Recurrence without a useful context helped less and only early. No-shape bursts scored 0.80 nats [0.47, 1.13] more than no-shape spaced nouns in Experiment 2 (pre-registered), but only at 100 exposures; at 1,000 the gain was 0.05 [−0.30, 0.40] (Experiment 3: 0.20). Fixed-shape bursts also copied more than fixed-shape spaced nouns, but at 100 exposures bursts left the stored shape weaker (stored preference 3.0 against 5.1 to 5.4 nats), so this registered comparison is not informative; at 1,000, with equal stored preference, bursts added about 0.5 nats.

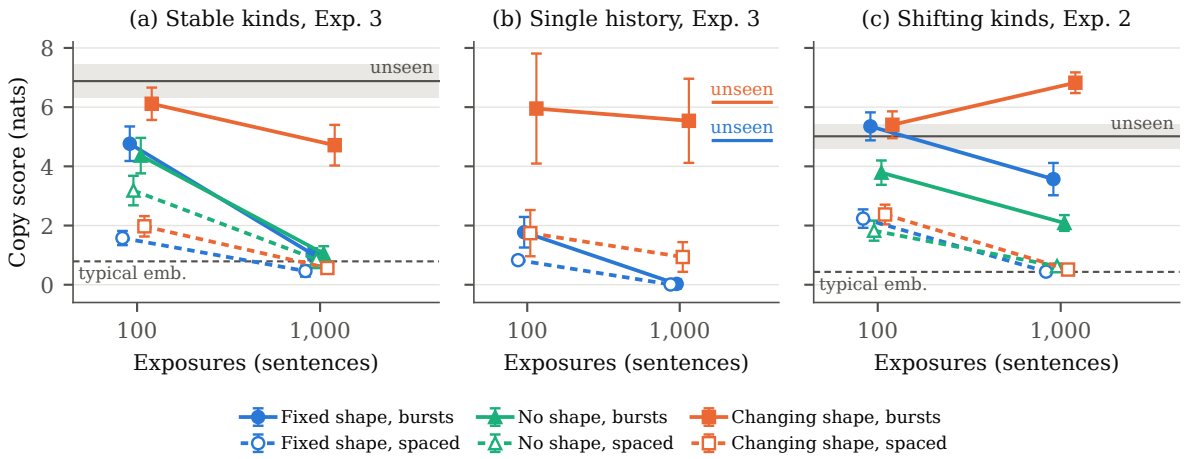


Figure 3: Copy score by history (colour) and presentation (solid: bursts; dashed: spaced) at 100 and 1,000 exposures (means over seeds, sizes averaged; 95% intervals; points offset for legibility). (a) Stable-kind corpora, Experiment 3 (ten new seeds). (b) Single-history corpora, Experiment 3 (five seeds): all familiar nouns have changing shapes (orange) or all have fixed shapes (blue). (c) Shifting-kind corpora, Experiment 2 (exploratory). Solid grey line and band: unseen nouns; dashed line: unseen nouns with a typical embedding. Experiment 2's stable-kind results are in Figure E2.

4.4 A stored shape lowers copying slightly but wins in prediction

Fixed- and changing-shape spaced nouns share frame, presentation and expected word class, and differ only in whether a shape can be stored. Fixed-shape spaced nouns copied 0.25 nats [0.15, 0.35] less (pre-registered), mostly at 100 exposures (0.40) rather than 1,000 (0.10). The difference was small but reliable, and not shown to be within ± 0.3 nats; Experiment 1 found 0.30 at 1,000 exposures and Experiment 2 none (-0.02). The copy score leaves out the stored shape, so it shows only whether the stored shape pushes down the copied one; a stored shape that simply raised its own probability would not show in it. Which shape the model finally predicts is another matter: fixed-shape spaced nouns preferred their own shape to the context's by 5.84 nats [5.57, 6.11] (pre-registered), 4.1 at 100 exposures and 7.6 at 1,000. Spaced nouns copied little in any case (under 2 nats), so the test is limited to weak copying. Fixed-shape spaced nouns also scored about 1 nat below no-shape spaced nouns (-1.02, pre-registered, and -1.00), but no-shape nouns expect a colour where the others expect a shape, so this difference reflects word class.

4.5 History sets whether the model copies, not what it copies

If changing-shape bursts taught the model to bind a noun to its current shape, the control, in which the query names another noun, should score lower. It did not. For changing-shape bursts at 1,000 exposures the context about the queried noun added 0.08 nats [0.02, 0.14] over the control, within ± 0.3 (pre-registered), and no other condition in mixed or fixed-only stable-kind corpora exceeded 0.14. The model copied the most recent shape after the queried noun, whichever noun the preceding sentence described, which training never required it to check. In changing-only corpora, however, changing-shape bursts showed a noun-specific part of 0.33 nats [0.18, 0.48], so when every new noun's context is useful, some binding to the noun appears. To see whose history matters, we paired a context about a noun of one history with a query naming another noun. At 1,000 exposures, a queried noun with changing-shape bursts copied 1.72 nats [1.36, 2.08] more than one with fixed-shape bursts. The same contrast for the context noun added only 0.47 [0.36, 0.58]. The difference, 1.25 [0.94, 1.56], was pre-registered.

4.6 Copying spreads between nouns

Because copying is not tied to the noun, what changing-shape bursts teach may reach other nouns. In Experiment 3's single-history corpora no noun of another history was present (Figure 3b). There, changing-shape bursts copied 5.51 nats [4.24, 6.78] more than fixed-shape bursts at 1,000 exposures, at all three sizes (pre-registered), so the main difference does not need nouns of the other history.

It was larger than in mixed corpora with the same seeds (4.27), mainly because fixed-shape bursts lost all their copying when no noun of another history shared the corpus: 0.03 nats, against 1.03 in mixed corpora, a difference of 1.00 [0.80, 1.20] (pre-registered). For changing-shape bursts this comparison was inconclusive. Unseen nouns also copied less in fixed-only than in mixed corpora with the same seeds (4.87 against 7.35), with changing-only corpora in between (6.16).

4.7 How the embedding sets copying

A noun enters the model only through its embedding, so whatever its history taught must be carried there. Could changing-shape bursts simply leave the embedding less trained? For every familiar noun we measured the embedding's cosine with its initial value, its distance from the trained kinds' mean and its norm, and regressed copy scores on them within each model. At 1,000 exposures changing-shape bursts copied 3.67 nats [3.03, 4.32] more than these measures predict (pre-registered), although their embeddings were only slightly closer to their initial values than those of fixed-shape bursts (cosine 0.85 against 0.78). At 100 exposures geometry still accounted for their copying (0.45 [−0.25, 1.15]). At 100 exposures, then, burst nouns' copying is accounted for by embeddings left near their initial state; by 1,000, changing-shape bursts copy more than their embeddings' geometry explains.

Three exploratory analyses of the Experiment 3 models locate it. First, it lies along a direction shared by nouns: adding to a noun's embedding the difference between the mean embeddings of changing- and fixed-shape bursts, taken from other nouns, raised copying at 1,000 exposures by 1.51 nats [1.20, 1.82] for fixed-shape spaced nouns and 1.99 for unseen nouns with a typical embedding, while a random vector of the same length changed it by at most 0.02 for these nouns (Table F4). The effect was largest at 3.4M (2.41 for fixed-shape spaced nouns) and smaller at 25.6M (1.00). Second, the noun's history is read early. Patching the queried noun's hidden state between a changing- and a fixed-shape burst noun (Zhang and Nanda, 2024) showed that after the second block the noun's own position no longer mattered and the effect had moved to the position where the shape is predicted. Third, first-layer attention from that position to the context's shape word tracked copying across conditions within each model (mean Spearman $\rho = 0.80$ [0.76, 0.85]). Changing-shape bursts thus move the embedding along a direction that goes with more first-layer attention to the latest shape.

This learned copying is narrow. With one unrelated sentence between context and query it almost vanished (at most 0.44 nats). With the context in another frame (*Look at the noun. It is shape and colour.*) changing-shape bursts kept 35 to 40% of it at 1,000 exposures. For colour, which varies freely within every kind, no noun copied.

4.8 When the whole corpus rewards copying

In Experiment 2's shifting-kind corpora, where the trained kinds' shapes can come only from context, every burst history kept copying (Figure 3c): at 1,000 exposures fixed-shape bursts scored 3.57 nats, no-shape bursts 2.09 and changing-shape bursts 6.82, more than unseen nouns (5.02). When the corpus rewards copying, recurrence alone sustains it.

4.9 Model size

In Experiment 3, the advantage of changing- over fixed-shape bursts at 1,000 exposures was 6.36, 2.44 and 2.24 nats at 3.4M, 10.9M and 25.6M parameters, while unseen nouns with a typical embedding copied more as models grew (−0.19, 1.20 and 1.37). In ten models of 85.6M parameters the advantage was 4.18 nats [2.95, 5.41] (pre-registered); changing-shape bursts still copied 4.85 nats after 1,000 exposures, against at most 1.2 for every other familiar history. The advantage was thus larger at 85.6M than at 25.6M, but these models also had a longer warm-up and other hardware, so size is not their only difference.

4.10 Pretrained models

We registered three predictions for Pythia models from 1B to 12B parameters (Biderman et al., 2023), using 22 one-token nouns with a typical colour (*sky*) and 22 without (*truck*), matched on frequency in the Pile (Gao et al., 2020; Liu et al., 2024): replacing a noun's embedding with a random one of untrained scale would raise colour copying, more than a typical embedding would, and colour-typical nouns would copy less. The models had stored typical colours, and real nouns copied 1.9 to 2.7 nats of colour. None of the predictions held (pre-registered): replacements changed copying by up to 0.6 nats in either direction, inconsistently across models, and most changes were not small enough to count as none (Appendix E, Figure E3).

5 Discussion

What decides copying. In stable-kind corpora, whether the model copies the latest shape after a noun depends on the noun's embedding. The tendency starts high for embeddings far from the trained nouns', which is why unseen nouns copy. Training removes it, by 1,000 exposures, unless the noun's own earlier sentences were the only source of its current shape. This held in new seeds, at 85.6M parameters and in corpora without nouns of the other history. Memory mattered, and its effect appears to arise in training rather than at test: where a shape could be stored, the context was not needed, and copying was lost. At test a stored shape lowered copying by 0 to 0.4 nats across three experiments, about a tenth of the effect of a useful context, and acted mainly by winning in prediction. The useful-context and memory accounts are thus complementary: in stable-kind corpora, after enough exposures, the model kept copying for a word only where memory could not do the job.

What this adds. Chan et al. (2022a) showed that bursty items whose meaning changes favour in-context learning across a population. We find the same rule acting on single words within one model, through each word's embedding. Two results go beyond it. Recurrence alone did not keep copying: no-shape bursts lost their advantage by 1,000 exposures. And copying learned for some words spread to others, so a word's tendency to copy can depend on the company it keeps in training. When the corpus as a whole rewards copying, recurrence of any kind sustains it.

Implications. In small models trained from scratch, tied or untied, probes with invented words can measure an unseen token's embedding more than anything learned about context; in Pythia this did not hold. Natural text resembles changing-shape bursts in one respect. In WikiText-103 (Merity et al., 2017), the 44 nouns of the pretrained test recurred within 128 words after a median 40% (colour-typical) and 24% (colour-variable) of their mentions, over 100 times more often than if spread evenly. A colour named before a noun twice in one article was usually the same (median agreement 0.98 and 0.86), whereas two mentions from anywhere agreed less often (0.46 and 0.37; Appendix F). Many such pairs are names (*Black Sea*), and our models' copying reached only the next sentence, so this is suggestive only.

Analogies with children. Two findings echo children's word learning, as analogies only. Changing-shape bursts resemble the boxes of Dewar and Xu (2010), uniform within and varying between; after one draw from a new box, 9-month-olds expected the next to match, an overhypothesis in the sense of Kemp et al. (2007). And spaced nouns first stored their shape better, while bursts kept the context in use; children retain words better from spaced exposures (Childers and Tomasello, 2002; Vlach et al., 2008) but learn labels repeated in successive sentences (Schwab and Lew-Williams, 2016).

Limitations

The language is artificial, with 32 kinds, three features and five sentence frames; the models are small (at most 85.6M parameters) and trained with one recipe, and most effects shrank from 3.4M to 25.6M parameters. Experiments 2 and 3 tested 100 and 1,000 exposures, and over training exposure is confounded with training time. The learned copying almost vanished beyond one sentence, transferred only partly to another frame, and did not apply to colour. No-shape comparisons are confounded with expected word class, and bursts weakened stored shapes at 100 exposures. The test of memory competition rests on spaced nouns, which copied little. Single-history corpora used five seeds. The tied-embedding check used one size, and it and the mechanism analyses were added after the results. The mechanism is located only coarsely; the heads involved were not identified, and a registered search for induction-like heads in Experiment 1 found none (Figure D1). Planned power for a 1-nat difference in Experiment 1 was 0.57. The pretrained test used one model family, 22 noun pairs and colour only. Earlier experiments in this language, not reported here, were registered in stages (Appendix C).

Ethics statement

All training data are artificial; the pretrained models are public. No data were collected from people. Code, corpora, results and all trained models will be released; anonymised copies are provided as supplementary material.

Use of AI assistants

An AI assistant (Claude, Anthropic) was used to help design analyses, write code, review code and design, draft and edit text and check citations. It

also reviewed the write-up of Experiment 1, which identified the two differences addressed by Experiment 2, and coded the colour typicality of the pretrained-model nouns in two independent passes. The authors are responsible for all content, including the accuracy of every citation.

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A Language and corpora

Language. Each corpus describes 32 kinds of object, each named by an invented noun, with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each an invented word from a list of ten, and is described by one sentence in one of five frames:

1. *A wug is a [shape] [colour] [texture] thing.*
2. *Look at the wug. It is [shape] and [colour].*
3. *Here is another wug. This one is [shape].*
4. *The wug I showed you is [shape], but a different [colour].*
5. *That is a wug because it is [shape].*

Shape is named in all five frames, colour in three and texture in one. The 448 sentences of a corpus make up one pass. Every word is one token of a byte-pair tokeniser with 329 entries, shared by all models.

Order and kinds. Within each pass the sentences about each kind are shuffled, cut into runs of four and the runs shuffled across kinds (grouped order). In stable-kind corpora each kind keeps one shape across its exemplars, while colour and texture vary. In shifting-kind corpora each kind receives a new shape, drawn at random, in every pass and keeps it within the pass. Experiment 1 also used stable-kind corpora in random order, with every pass shuffled.

Text read. Each model read fresh text once, in non-overlapping 128-token windows in a fixed order: 160,000 windows for the 3.4M, 25.6M and 85.6M models and 256,000 for the 10.9M models, which share the rules and nouns of the shorter corpus. Each pass is arranged with its own random draw.

New nouns. Forty new nouns per corpus, three per condition and four unseen, rotated over nouns across seeds. A noun's sentences (its exposures) were placed evenly between 5% and 95% of the windows read, at most one unit (a sentence, or a burst of four) per window, between the trained kinds' runs, at a random position in the window. In Experiment 1 fixed- and changing-shape nouns appeared in frames 1 to 5 and no-shape nouns in a sixth frame, *The [noun] is [colour] and [texture]*, which the trained kinds also used in one sentence in fifteen. In Experiments 2 and 3 fixed- and changing-shape nouns appeared only in frame 1 and no-shape nouns only in *A [noun] is a [colour] [texture] thing*, which the trained kinds used in one sentence in fifteen. A checker re-read every corpus and verified exposures, units, frames, shapes within bursts and window placement; all corpora matched checksums recorded before training.

B Models and training

Decoder-only transformers with pre-layer normalisation, learned position embeddings, GELU feed-forward layers, AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up then cosine decay, and dropout 0.1 (Table B1). Input and output embeddings are untied: the output layer starts as a copy of the input embedding and is trained separately, and every final model was checked to have different input and output matrices. We chose untied embeddings after finding that models of an earlier study in this language, meant to be tied, had been trained untied because the tie was lost on the graphics hardware used; their results are not reported here. The ten tied-embedding models of Appendix F were trained on other hardware, with input and output embeddings sharing one matrix; the first was checked after training and its matrices were identical. Models of Experiments 1 and 2 and the three smaller sizes of Experiment 3 were trained on one AMD graphics card (DirectML); the 85.6M and tied models on an Apple M3 Ultra.

Table B1: Model and training settings. Batches are of 128-token windows.

Setting	3.4M	10.9M	25.6M	85.6M
Layers, heads, width	4, 4, 256	6, 8, 384	8, 8, 512	12, 12, 768
Feed-forward width	1,024	1,536	2,048	3,072
Steps	5,000	8,000	10,000	10,000
Batch	32	32	16	16
Peak learning rate	3×10^{-4}	2×10^{-4}	1×10^{-4}	1×10^{-4}
Warm-up steps	200	400	500	1,000

Data and code. All code, corpora, test items, scores and analysis outputs, and the final weights and training logs of every model, will be released on publication; anonymised copies of the code and results are in the supplementary material, with the command that regenerates each table and figure.

C Registrations, terms and registered tests

Registrations. All registrations are in one public project on the Open Science Framework (link withheld for review). The project began with earlier studies of this language (a set of training experiments, an analysis of natural corpora and a test of pretrained models), registered in stages, some after related results were known; they are not reported here. During that work we found that the earlier models' embeddings had not been tied as intended (Appendix B) and that a loading error had changed their scores; both were corrected and disclosed in the registrations. The three experiments of this paper were registered as amendments. Experiment 1 was registered before its corpora were built. Experiment 2, with the row analysis and the pretrained test, was registered after Experiment 1's results and after its corpora were built, before any of its models was trained or any pretrained model was run on its items. Experiment 3 was registered after all of Experiment 2's results, including the exploratory analyses that it tests, and before any of its models was trained (Appendix F). Each registration states what was known when it was written. The registrations use other names for the conditions (Table C1).

Table C1: Main-text terms and registration terms, as used in the registrations and in Appendices D to F.

Main text	Registrations
Experiment 1	Amendment 9 (the familiarity study)
Experiment 2	Amendment 10, Part A
Embedding analysis of Experiment 1 (Section 4.1)	Amendment 10, Part C
Pretrained test (Section 4.10)	Amendment 10, Part B
Experiment 3	Amendment 11 (Parts A, B and C)
Stable-kind corpora	Regular (grouped) corpora
Shifting-kind corpora	Reassigned (grouped) corpora
Single-history corpora (changing-only, fixed-only)	pure_changing, pure_fixed
Spaced, bursts	singles, runs
Changing shape	uninformative (Experiment 1), rotating (Experiments 2 and 3)
No shape	shape-absent, absent
Copy score	carry
Stored preference	knowledge
Embedding	row
Typical embedding	typical row
Random embedding of untrained scale	atypical_init

Registered tests. Table C2 lists every registered hypothesis of the three experiments, the embedding analysis and the pretrained test, with its outcome under the registered decision rules (Holm correction within each registered family; direction in at least two of three model sizes; for the pretrained test, Holm within each model and support in at least four of five models). Equivalence bounds were set in each registration: ± 1 nat in Experiment 1 (about one seventh of the fall from unseen to typical embeddings in earlier work in this language); in Experiment 2, ± 0.3 nats for H40, H41 and H44 (the smallest history effect in Experiment 1), ± 1 nat for the check K1, and the larger of 0.3 nats and one fifth of a pilot effect for H48; in Experiment 3, ± 0.3 nats for H52, H54 and H59.

Table C2: Registered hypotheses and outcomes. Full statistics: Tables D1 (Experiment 1), E1 (Experiment 2), E6 (row analysis), E7 (pretrained models) and F1 (Experiment 3).

Label	Prediction (main-text terms)	Outcome
Experiment 1 (Amendment 9)		
M1	Fixed-shape nouns store their shape (stored preference > 0.2 nats at 1,000 exposures, rising)	Passed in all three corpora
H30	Copying falls faster with exposure for fixed-shape than no-shape nouns	Not supported (wrong direction)
H31	No-shape nouns (1,000 and 10,000) differ from unseen nouns with a typical embedding; equivalence ± 1	Different (0.33) and equivalent
H32	Fixed-shape nouns' preference for their own shape, measured in the copy items, rises with exposure	Supported
H33	Ablating candidate heads removes more than half of unseen nouns' copying, more than random heads	Not supported
H34	Ablating candidate heads lowers trained kinds' copying in shifting-kind corpora	Supported; random heads cut it nearly as much
H35	Candidate heads' attention falls faster for fixed-shape nouns	Not tested (required H33)
R1	Unseen nouns copy more in grouped than random order	Replicated
Experiment 2 (Amendment 10, Part A); stable-kind corpora		
M2	Unseen nouns copy more than 1 nat	Passed
K1	Fixed-shape nouns' stored preference equal (± 1) in bursts and spaced	Not equivalent: H41 not interpreted
H40	No-shape: bursts differ from spaced; equivalence ± 0.3	Bursts higher
H41	Fixed shape: bursts differ from spaced; equivalence ± 0.3	Bursts higher (not interpreted)
H42	Changing shape: bursts copy more than spaced	Supported
H43	Spaced: fixed shape copies less than no shape	Supported; confounded by expected word class
H44	Spaced no-shape nouns differ from unseen nouns with a typical embedding; equivalence ± 0.3	Higher; not equivalent
Experiment 3 (Amendment 11); stable-kind corpora; Holm over H50–H53, H55, H56		
M3	Unseen nouns copy more than 1 nat	Passed
H50	At 1,000: changing-shape bursts copy more than fixed-shape bursts (primary)	Supported (3.68)
H51	Changing shape: bursts copy more than spaced	Supported (4.14)
H52	Spaced: fixed differs from changing shape; equivalence ± 0.3	Fixed lower (-0.25); not equivalent
H53	Fixed-shape spaced nouns prefer their own shape to the context's	Supported (5.84)
H54	Noun-specific copying of changing-shape bursts within ± 0.3	Within ± 0.3 (0.08)
H55	Changing-shape bursts copy more than embedding geometry predicts	Supported (3.67)
H56	The queried noun's history matters more than the context noun's	Supported (1.25)
M3B, H57	85.6M models: unseen nouns copy; changing-shape bursts copy more than fixed-shape bursts	Passed; supported (4.18)
M3C	Single-history corpora: unseen nouns copy	Passed
H58	Changing-shape bursts (changing-only) copy more than fixed-shape bursts (fixed-only)	Supported (5.51)
H59	Single-history minus mixed corpora, same history; equivalence ± 0.3	Changing bursts inconclusive; fixed bursts lower (-1.00)
Row analysis (Amendment 10, Part C); Experiment 1 models		
H49	Scaling unseen nouns' embeddings to the trained kinds' norm lowers copying	Supported (driven by the 3.4M models)
Pretrained models (Amendment 10, Part B)		
H45	A random embedding of untrained scale raises copying over the own embedding	Not supported (2 of 5 models)
H46	The same embedding raises copying over a typical embedding	Not supported (2 of 5 models)
H47	Colour-typical nouns copy less than colour-variable nouns	Not supported (0 of 5)
H48	The H45 increase differs between colour-typical and colour-variable nouns	Inconclusive

D Experiment 1 (Amendment 9)

Registration. The design, hypotheses and analysis were registered before any corpus was built (Amendment 9). An addendum, written the same day before any corpus was built, fixed details the registration left open (36 context draws per noun; uniform frame proportions; exposures placed at evenly spaced read positions from 5% to 95% of training, after warm-up for every size; random in-window position; the 40 nonce nouns; how the pilot fixes the number of nouns per cell; how a hypothesis counts as supported; the gate for H35; and that heads without prefix matching are described as copying heads). A pilot (one seed, 3.4M models, never used in a test) set three nouns per cell; with the registered rule the expected power for a 1-nat difference was 0.57, below the planned 0.8, which we report as a shortfall. Corpus checksums were recorded before training and matched on the training machine; a checker verified every exposure, window and frame from the tokens. All models were trained on one machine.

Design. In every model, 40 nonce nouns fill 13 cells (three per cell, four unseen): fixed shape, uninformative shape (a new shape in every sentence, drawn from the corpus's shape distribution) and shape-absent (a sixth frame, *The noun is colour and texture*, also used by trained kinds in one sentence in fifteen), each at 10, 100, 1,000 and 10,000 exposures, plus unseen. Nouns rotate over cells across ten seeds. Each familiarity sentence lies wholly inside one 128-token window, at most one per window, between runs in grouped corpora. Base corpora: Regular grouped, Regular random, Reassigned grouped; sizes 3.4M, 10.9M and 25.6M; ten seeds; 90 models with untied embeddings. Measures are read from the full log-softmax over the ten shape words at the position that predicts the shape in *A noun is a*: carry $= L(s) - \bar{L}$ after a context sentence with shape s; own-shape preference $= L(s^*) - \bar{L}$; knowledge $= L(s^*)$ minus the mean over the other nine shapes after an unrelated sentence, where $\bar{L}$ is the mean over shapes other than s and s^*.

Table D1: Registered tests of the familiarity study (Regular grouped models unless stated; seed-level estimates over ten seeds, sizes averaged within seed; nats, proportions for H33a, attention probability for H35). Holm p is adjusted over H30 to H34; H35 was not entered because H33(a) failed. Sizes: number of the three model sizes with the registered direction.

	Test	Estimate	95% CI	Holm p	Sizes	Verdict
M1	Knowledge at 1,000 exposures, fixed shape (each corpus)	9.53; 8.61; 8.26				Passed
H30	Carry falls faster with exposure for fixed-shape than shape-absent nouns (slope difference)	0.16	[0.01, 0.30]	1.000	0	Not supported
H31	Shape-absent (1k and 10k) minus unseen with typical embedding (two-sided; TOST ± 1)	0.33	[0.13, 0.52]	.013	3	Different, and equivalent within 1 nat
H32	Own-shape preference rises with exposure, fixed shape (slope)	2.97	[2.79, 3.16]	$<.001$	3	Supported
H33a	Ablating candidates removes more than half of unseen carry (proportion)	0.22	[0.07, 0.36]	1.000	2	Not supported
H33b	Random heads minus candidates, unseen carry after ablation	0.64	[–0.43, 1.71]	1.000	2	Not supported
H34	Reassigned: trained-kind carry intact minus candidates ablated	7.41	[6.02, 8.80]	$<.001$	3	Supported
H35	Attention slope difference, fixed shape minus shape-absent	0.002	[0.000, 0.005]	–	2	Not tested (gate)
R1	Replication: grouped minus random Regular, unseen carry	4.68	[3.71, 5.65]	$<.001^a$	3	Replicated

[a] Outside the Holm family (replication); unadjusted one-sided p. M1 values: Regular grouped; Regular random; Reassigned grouped (all with slopes above zero, $p < .001$). H31 TOST $p < .001$.

Table D2: Carry (nats) by noun type and exposure, mean of seed values (sizes averaged). Unseen: never in training; typical: the unseen nouns with a typical embedding; trained: the 32 base kinds.

Corpus	Unseen	Typical	Trained	Familiar nouns	10	100	1k	10k
Regular grouped	4.53	−0.33	−0.02	Fixed shape	3.70	0.30	−0.30	−0.37
				Uninformative	3.57	0.16	0.00	−0.01
				Shape-absent	4.47	0.86	0.05	−0.05
Reassigned grouped	6.85	1.57	9.98	Fixed shape	6.40	2.13	0.70	0.60
				Uninformative	6.34	2.53	0.30	0.15
				Shape-absent	6.97	4.89	2.04	1.86
Regular random	−0.15	−0.72	−0.10	Fixed shape	−0.13	−0.13	−0.06	−0.05
				Uninformative	−0.10	−0.03	−0.02	−0.01
				Shape-absent	−0.19	−0.09	−0.07	−0.05

Table D3: Knowledge and own-shape preference (nats) in Regular grouped models, mean of seed values.

Type	Measure	10	100	1k	10k
Fixed shape	Knowledge	1.59	8.02	9.53	10.61
	Own shape	1.41	7.74	9.56	10.71
Uninformative	Knowledge	0.04	−0.03	0.00	0.03
	Own shape	0.01	−0.03	−0.01	0.03
Shape-absent	Knowledge	−0.14	−0.17	0.12	0.08
	Own shape	0.11	−0.16	0.10	0.06

Table D4: Head analysis by corpus and size (means over ten seeds). Candidates were the k heads with the highest product of prefix-matching and copying scores; in grouped models they were in layer 0 in all but one model (a 3.4M Regular grouped model chose one layer-1 head), and in random-order models a few were in later layers. In the 3.4M and 10.9M models layer 0 has exactly k non-candidate heads, so the random draws repeat one set. Prefix: highest prefix-matching score of any head (uniform attention gives about 0.027). Carry of unseen nouns and of trained kinds intact, with the candidates mean-ablated, and with random heads ablated (mean of 200 or 20 draws). DLA: change in unseen carry predicted by the candidates' direct paths.

				Unseen carry			Trained carry			
Corpus	Size	k	Prefix	Intact	Cand.	Random	Intact	Cand.	Random	DLA
Regular grouped	3.4M	2	0.028	8.10	−0.42	5.26	−0.25	−0.85	−0.71	−0.04
	10.9M	4	0.037	2.98	5.80	1.95	0.08	0.17	−0.40	−0.01
	25.6M	6	0.037	2.50	0.73	0.81	0.12	−0.60	−0.18	−0.01
Reassigned grouped	3.4M	2	0.029	6.48	4.22	3.27	9.59	2.83	3.37	−0.12
	10.9M	4	0.032	7.21	6.56	4.82	10.10	5.02	2.45	−0.03
	25.6M	6	0.031	6.88	0.26	3.26	10.26	−0.11	3.39	−0.01
Regular random	3.4M	2	0.028	−0.24	−0.09	−0.34	−0.08	0.00	−0.11	−0.01
	10.9M	4	0.030	0.04	0.25	0.03	−0.06	0.04	−0.11	−0.01
	25.6M	6	0.030	−0.25	0.00	−0.37	−0.18	−0.10	−0.27	−0.01

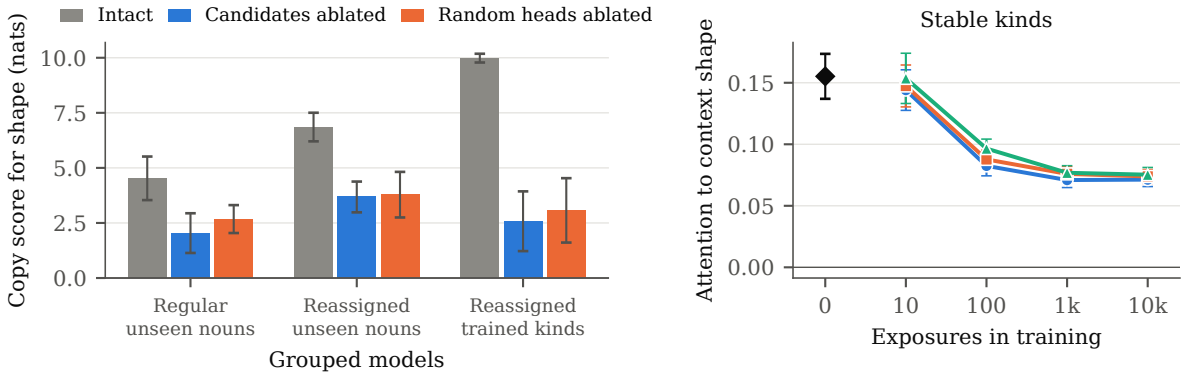


Figure D1: Experiment 1. Left: copy score intact, with the candidate heads mean-ablated, and with random heads ablated (nats; mean over ten seeds and three sizes, 95% intervals over seeds). Right: the candidates' attention from the position that predicts the shape to the context's shape word, stable-kind models (exploratory).

E Experiment 2, the embedding analysis and the pretrained test (Amendment 10)

Registration. Amendment 10 has three parts: Part A is Experiment 2, Part B the pretrained test and Part C the row analysis. It was registered after the Amendment 9 results, after its corpora were built and checksummed, and before any of them was used, any of its models was trained, or any pretrained model was run on its items. It answered two confounds raised by a blind review of the Amendment 9 results (by an AI assistant given only the write-up of Experiment 1): familiarity nouns never recurred within a window, and shape-absent nouns never appeared in a frame with a shape slot after *is a*. The noun lists for Part B were fixed, with checksums, before the pilot. A pilot on Pythia-410M (never in a test) gave a mean increase of 0.13 nats from the original to the atypical_init row, so the smallest effect of interest for H48 was 0.30 nats by the registered rule. One deviation: some Pythia tokenizers define no padding token, so the scoring script falls back to token id 0; padded positions are masked and only the last real token is read, and batched and single-item scores agreed exactly. No score had been produced by the affected models.

Part A design. Everything not stated is as in Amendment 9 (Appendix D). Histories: *fixed*, frame 1 (*A noun is a shape colour texture thing*) with the noun's own shape; *absent*, frame 7 (*A noun is a colour texture thing*), which names no shape; *rotating*, frame 1 with a shape drawn from the corpus's shape distribution, a new one for every sentence when presented singly and one per run, redrawn for each run, when presented in runs. Presentation: *single*, one sentence per unit; *run*, four consecutive sentences about the noun in one window. Exposures (100 or 1,000) count sentences, so runs and singles are matched in sentences, not in units or spacing. Trained kinds use frame 7 in one sentence in fifteen, so the frame is not new. Three nouns per cell (12 cells) and four unseen nouns; Regular grouped (confirmatory) and Reassigned grouped (exploratory) base corpora; three sizes and ten seeds (60 models). A checker re-read every corpus and verified exposures, units, frames, shapes within runs and window placement; all 40 corpora matched checksums recorded before training.

Table E1: Registered tests of Part A (Regular grouped models; 100 and 1,000 exposures pooled; seed-level estimates over ten seeds, sizes averaged within seed; nats). Holm p is adjusted over H40 to H44. Sizes: number of the three sizes with the estimate's direction. Readings follow the registered table: "above" means the Holm threshold was met and the direction held at two of three sizes; TOST p for H40, H41 and H44 was above .99, so none was equivalent.

	Test	Estimate	95% CI	Holm p	Sizes	Reading
M2	Unseen carry minus 1 nat (manipulation check)	5.35	[4.59, 6.12]	$<.001^a$		Passed
K1	Knowledge, fixed runs minus fixed singles (equivalence ±1)	−1.09	[−1.45, −0.73]	$<.001^a$		Not equivalent (H41 not interpreted)
H40	Absent runs minus absent singles (two-sided; TOST ±0.3)	0.80	[0.47, 1.13]	$<.001$	3	Above
H41	Fixed runs minus fixed singles (two-sided; TOST ±0.3)	1.69	[1.29, 2.08]	$<.001$	3	Above (not interpreted)
H42	Rotating runs minus rotating singles (one-sided)	3.39	[2.69, 4.09]	$<.001$	3	Supported
H43	Fixed singles minus absent singles (one-sided)	−1.02	[−1.27, −0.76]	$<.001$	3	Supported
H44	Absent singles minus unseen with typical row (two-sided; TOST ±0.3)	1.42	[1.12, 1.73]	$<.001$	3	Above

[a] Outside the Holm family; unadjusted p. The M2 estimate is unseen carry minus 1 nat, so unseen carry was 6.35 nats.

Reanalysis of the Experiment 2 models (exploratory, added after the results). For every familiar noun in the 60 final models we recorded its input row's cosine with its initial row (rebuilt from the seed; the rebuild reproduced the unseen nouns' rows, which receive no gradient, in every model), its distance from the trained kinds' mean row and its norm; the copy score with its own row, with a typical row and with the row of a noun of another history (rows swapped between changing- and fixed-shape bursts and between changing-shape bursts and spaced nouns); the copy score with one sentence about a trained kind between context and query (gap), with the context in the frame *Look at the noun. It is shape and colour.* (other frame), and for the context's colour after *A noun is a shape* (colour); and first-layer attention from the query position to the context's shape, averaged over heads. Full results, including shifting-kind corpora

Table E2: Carry (nats) by cell, mean of seed values (sizes averaged) with 95% intervals over ten seeds. Unseen nouns have no exposures; the typical row replaces an unseen noun's input row with one typical of trained nouns.

Corpora	Noun	100 exposures	1,000 exposures
Stable kinds	Fixed shape, spaced	1.52 [1.18, 1.86]	0.52 [0.36, 0.69]
	Fixed shape, bursts	4.43 [3.49, 5.36]	0.99 [0.68, 1.30]
	No shape, spaced	3.09 [2.56, 3.63]	0.98 [0.76, 1.20]
	No shape, bursts	4.65 [3.94, 5.36]	1.03 [0.59, 1.47]
	Changing shape, spaced	1.59 [1.30, 1.88]	0.50 [0.38, 0.62]
	Changing shape, bursts	5.02 [4.18, 5.86]	3.85 [2.93, 4.77]
	Unseen	6.35 [5.59, 7.12]	
	Unseen, typical embedding	0.61 [0.28, 0.95]	
Shifting kinds	Fixed shape, spaced	2.23 [1.92, 2.55]	0.44 [0.37, 0.51]
	Fixed shape, bursts	5.35 [4.88, 5.82]	3.57 [3.02, 4.11]
	No shape, spaced	1.83 [1.49, 2.17]	0.63 [0.53, 0.72]
	No shape, bursts	3.79 [3.38, 4.20]	2.09 [1.82, 2.35]
	Changing shape, spaced	2.37 [2.04, 2.71]	0.51 [0.38, 0.64]
	Changing shape, bursts	5.40 [4.95, 5.86]	6.82 [6.48, 7.17]
	Unseen	5.02 [4.61, 5.42]	
	Unseen, typical embedding	0.43 [0.32, 0.55]	

Table E3: Knowledge of the noun's own shape and expectation of a shape rather than a colour at the query position (slot: log of the summed probability of the ten shapes minus that of the ten colours), Regular grouped models, mean of seed values (exploratory). Unseen nouns: knowledge −0.04, slot 1.49.

	Knowledge		Slot	
Noun	100	1,000	100	1,000
Fixed shape, singles	5.37	7.80	4.03	6.16
Fixed shape, runs	3.02	7.98	3.28	6.24
Shape-absent, singles	0.00	−0.09	−2.98	−4.93
Shape-absent, runs	0.05	−0.25	−1.12	−4.43
Rotating shape, singles	−0.08	0.01	5.04	5.30
Rotating shape, runs	0.14	0.06	3.74	5.24

and 100 exposures, are in the supplementary material.

Table E4: Experiment 2 reanalysis, stable-kind corpora, 1,000 exposures (unseen nouns: no exposures). Copy scores in nats; cosine of the row with its initial value; first-layer attention from the query to the context's shape. Seed-level means. Swapping rows between changing- and fixed-shape bursts moved the copying with the row (2.88 nats [2.01, 3.76] in each direction). In a noun-level regression of the copy score on row geometry (cosine with the initial row, distance from the trained kinds' mean, norm, size and seed; standard errors clustered by seed), changing-shape bursts scored 3.19 nats [2.46, 3.91] more than geometry predicts at 1,000 exposures and −0.09 [−0.51, 0.33] at 100.

Noun	Own row	Typical row	Gap	Other frame	Colour	Cos. initial	Attention
Unseen	6.35		0.16	4.38	−0.30	1.000	0.186
Fixed, spaced	0.52	0.67	0.05	0.12	−0.10	0.732	0.089
Fixed, bursts	0.99	0.61	0.13	0.32	−0.12	0.778	0.099
No shape, spaced	0.98	0.63	−0.04	0.71	−0.22	0.793	0.102
No shape, bursts	1.03	0.68	−0.07	0.66	−0.34	0.806	0.097
Changing, spaced	0.50	0.61	0.10	0.20	−0.10	0.794	0.108
Changing, bursts	3.85	0.66	0.37	1.53	−0.12	0.849	0.168

Part C (embedding rows in the Amendment 9 models). In the 60 final grouped models of Amendment 9 we replaced the input rows of unseen nouns and scored carry again; the unmodified condition reproduced the Amendment 9 scores. Conditions: own row; own direction scaled to the mean norm of the trained kinds' rows (ownnorm); a random direction at that norm (randnorm); the trained kinds' mean row. Shape-absent nouns at 10,000 exposures were also scored with their own direction scaled to the unseen nouns' mean norm (shrunk).

Part B (pretrained models). 44 nouns in 22 pairs, matched on Pile frequency and count/mass type,

Table E5: Crossed histories (exploratory), stable-kind corpora, 1,000 exposures (unseen nouns: none): copy score when the context sentence is about a noun of one history (rows) and the query names a different noun of another (columns); 24 items per pair and model, seed-level means. Averaged over familiar histories, a queried noun with changing-shape bursts copied 1.35 nats [0.97, 1.74] more than one with fixed-shape bursts, and a context noun with changing-shape bursts raised copying by 0.39 [0.19, 0.58]. Full results are in the supplementary material.

Context \ query	Unseen	Fixed spaced	Fixed bursts	No shape spaced	No shape bursts	Changing spaced	Changing bursts
Unseen	6.34	0.60	0.73	1.16	1.01	0.42	2.61
Fixed, spaced	3.67	0.62	0.75	0.94	0.93	0.44	1.73
Fixed, bursts	5.14	0.68	1.01	1.07	1.27	0.53	2.45
No shape, spaced	4.10	0.53	0.74	1.12	1.13	0.39	1.67
No shape, bursts	4.30	0.62	0.91	0.89	1.01	0.43	2.07
Changing, spaced	4.03	0.59	0.91	1.26	1.12	0.50	1.86
Changing, bursts	6.49	0.85	1.15	1.49	1.39	0.66	3.80

Table E6: Part C: carry (nats) by input row, mean over seeds (sizes averaged). H49 (registered, Regular grouped): ownnorm minus own −0.52 [−0.85, −0.20], one-sided p = .003, direction at 2 of 3 sizes, supported. In Reassigned models the same contrast went the other way (0.27 [0.12, 0.41]; exploratory). Randnorm minus ownnorm: 0.09 (Regular), 0.03 (Reassigned).

	Unseen nouns				Shape-absent, 10k	
Corpus	Own	Ownnorm	Randnorm	Trained mean	Own	Shrunk
Regular grouped	4.53	4.00	4.09	−0.44	−0.05	0.53
Reassigned grouped	6.85	7.12	7.15	1.05	1.86	4.16

were coded for colour typicality by two independent AI passes with a fixed prompt before any model was run (exact agreement 94%, weighted $\kappa = 0.96$); nouns that are also colour terms or that both passes flagged as polysemous were excluded. Colour-typical nouns are mostly natural kinds, so kind was a covariate. Each noun was read in three neutral frames with each of the eleven basic colour terms that are one token; carry is the mean over frames and ordered colour pairs of $\log p(c_1 \mid c_1) - \log p(c_1 \mid c_2)$, with colour probabilities renormalised over the colour set. Only the noun's input row was changed.

Table E7: Part B registered tests by primary model (nouns or pairs as units; Holm within model over H45 to H48). H45: atypical_init minus original; H46: atypical_init minus typical row; H47: colour-typical minus colour-variable nouns, original rows; H48: difference between groups in the H45 increase (TOST against ±0.30). A hypothesis needed Holm in at least four of the five models. Outcome: H45 and H46 met Holm in two of five models, H47 in none, and H48 was inconclusive (equivalent in one model).

	H45		H46		H47		H48	
Model	Est.	Holm	Est.	Holm	Est.	Holm	Est.	TOST p
1B	0.20	met	0.21	met	0.00	–	0.07	.097
1.4B	0.34	met	−0.22	–	−0.01	–	−0.08	.021
2.8B	−0.38	–	0.08	–	−0.16	–	0.11	.222
6.9B	0.07	–	0.36	met	−0.09	–	0.10	.127
12B	−0.35	–	−0.21	–	0.11	–	0.08	.066

Mixed model over the five primary models (carry ∼ condition × group + model + kind; random intercepts for noun and pair, random condition slopes by noun): atypical_init 0.00 (p = .956), typical row −0.04 (p = .559), colour-variable group 0.06 (p = .527).

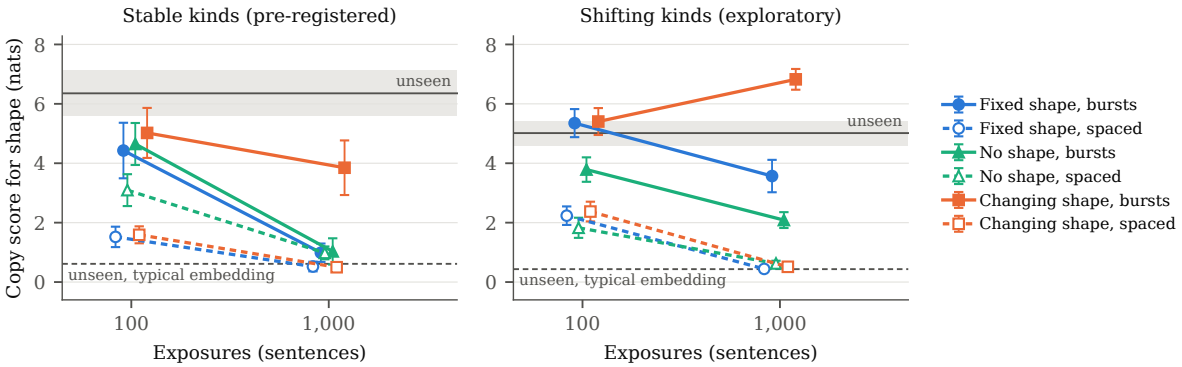


Figure E2: Experiment 2: copy score by history (colour) and presentation (solid: bursts; dashed: spaced) at 100 and 1,000 exposures, in stable-kind (left, pre-registered tests) and shifting-kind (right, exploratory) corpora (means over ten seeds, sizes averaged; 95% intervals). Solid line and band: unseen nouns; dashed line: unseen nouns with a typical embedding.

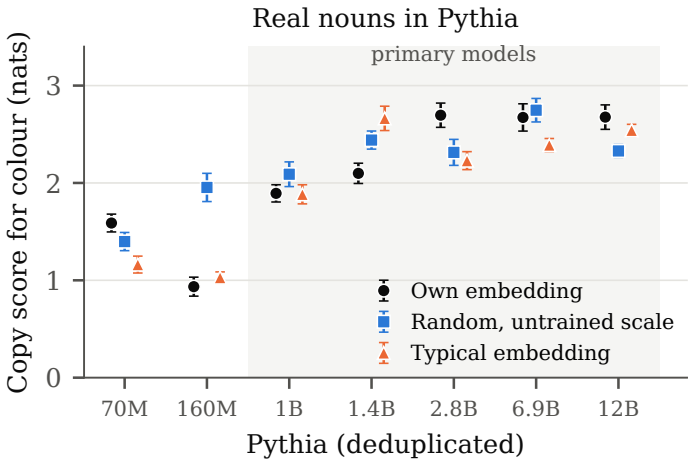


Figure E3: Copy score for colour of real nouns in Pythia models, with their own input embeddings or replaced ones (means over 44 nouns, 95% intervals). The registered tests used the 1B to 12B models; 70M and 160M are descriptive.

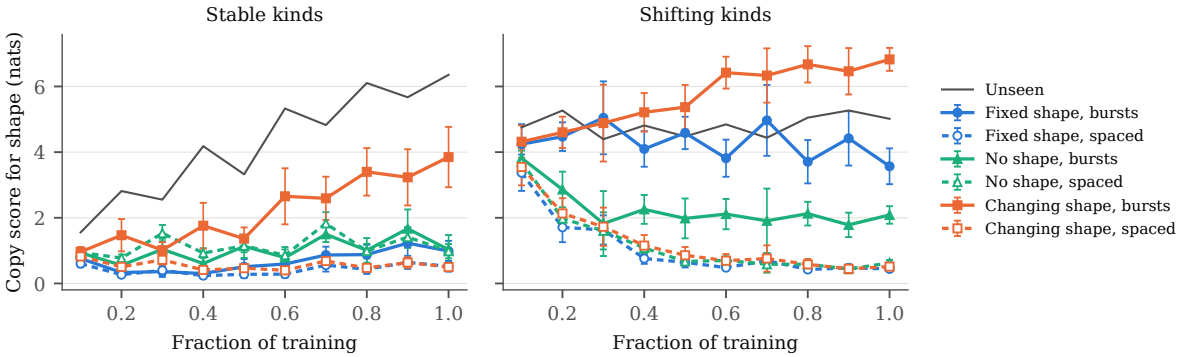


Figure E4: Experiment 2. Copy score over training for nouns with 1,000 exposures, by history and presentation, and for unseen nouns (exploratory; saved weights every 1,000 steps; means over ten seeds, sizes averaged; 95% intervals). A noun with 1,000 exposures has had about $1{,}000 \times (t - 0.05)/0.9$ sentences by fraction t of training, so exposure and training time rise together. Small zig-zags arise because the three sizes save weights at different fractions of training.

Table E8: Part B: carry of the context's colour (nats; mean over nouns) by model and input row, and with original rows by group. Atypical: the atypical_init direction at the noun's own norm (exploratory). 70M and 160M are descriptive. Prior entropy: entropy of the colour distribution after *The noun was* with no context (nats; uniform over eleven colours is 2.40).

Model	Original	Atypical_init	Atypical	Typical row	Colour-typical	Colour-variable	Prior entropy
70M	1.59	1.40	1.41	1.16	1.61	1.56	2.09
160M	0.93	1.95	1.95	1.03	0.87	1.00	2.01
1B	1.89	2.09	2.09	1.88	1.89	1.89	1.95
1.4B	2.10	2.44	2.44	2.66	2.10	2.10	1.97
2.8B	2.70	2.31	2.31	2.23	2.61	2.78	1.92
6.9B	2.67	2.75	2.75	2.39	2.63	2.72	1.94
12B	2.68	2.33	2.33	2.54	2.73	2.62	1.91

Table E9: Part B checks (exploratory). Stored colour: with no context, log probability of a colour-typical noun's coded colour minus the mean over the other colours after *The noun was* (nats; mean over 22 nouns, and the share of nouns above zero). TOST: equivalence of the H45 and H46 differences within ± 0.30 nats over nouns. Norm only: atypical minus atypical_init (same direction, different norm).

Model	Stored colour	Share > 0	H45 TOST p	H46 TOST p	Norm only
70M	0.72	0.73	.059	.129	0.014
160M	0.90	0.77	1.000	1.000	0.000
1B	1.28	0.82	.125	.091	0.001
1.4B	1.30	0.91	.736	.172	0.000
2.8B	1.55	0.86	.776	.001	0.000
6.9B	1.66	0.95	.012	.801	−0.002
12B	1.73	0.86	.740	.023	−0.001

F Experiment 3 (Amendment 11)

Registration. Amendment 11 is Experiment 3. It was registered in the project's public OSF record (link withheld for review) after all results of Experiment 2 and its exploratory analyses, after its corpora were built and checksummed, and before any of its models was trained (apart from deleted 100-step pipeline checks), to test the exploratory findings of Experiment 2 on new models. Part A repeated the stable-kind (Regular grouped) design of Experiment 2 with ten new seeds at 3.4M, 10.9M and 25.6M parameters (30 models). Part B trained ten models of 85.6M parameters (12 layers, 12 heads, width 768) on the same corpora as the 25.6M models. Part C trained five seeds at each of the three smaller sizes on corpora in which all 36 familiar nouns had changing shapes (pure_changing) or all had fixed shapes (pure_fixed), nine nouns per cell, with the same exposures and share of text as the mixed design (30 models). The assignment of noun forms to cells follows the Experiment 2 rule by seed position; model weights, reading order and placement are new. All 40 corpora passed a checker and matched checksums recorded before training. Registered rules for failed models (non-finite loss, or a final loss more than 0.5 nats above the median of its size and design) and for complete seeds were applied; no model failed. Part A used Holm correction over H50, H51, H52 (difference), H53, H55 and H56, with direction required at two of three sizes; equivalence tests used two one-sided tests against ± 0.3 nats. H55 is a seed-level test: within each model, the copy scores of the 18 familiar nouns at 1,000 exposures were regressed on the row's cosine with its initial value, its relative distance from the trained kinds' mean row, its norm and a changing-burst indicator, and the indicator's coefficient was averaged over sizes within seed. The 85.6M models were trained on an Apple M3 Ultra (MPS, PyTorch 2.14.1), the others on the machine used for Experiment 2. One deviation: after all 85.6M models had finished, their training loop was started again by mistake and stopped at step 1; only a configuration file was rewritten, with the same content, and no model or score changed.

Mechanism (exploratory, added after the results). Three analyses of the 30 mixed-design models (scripts/a11_mechanism.py). *Steering*: in each model, the direction v is the mean input row of changing-shape bursts minus that of fixed-shape bursts at one exposure level; it is added to the rows of other nouns and the registered items are scored. The direction from 1,000-exposure bursts is applied to spaced

Table F1: Registered tests of Experiment 3 (seed-level estimates, nats; sizes averaged within seed; t-based 95% intervals). Holm p: adjusted over the six Part A tests marked ^{h}; other p values are unadjusted. Sizes: number of the three sizes with the estimate's direction.

	Test	Estimate	95% CI	p	Sizes	Reading
M3	Unseen nouns' copy score minus 1 nat (manipulation check; one-sided)	5.88	[5.31, 6.45]	<.001		Passed
H50^h	At 1,000: changing-shape bursts minus fixed-shape bursts (one-sided; primary)	3.68	[2.99, 4.37]	<.001	3	Supported
H51^h	Changing shape: bursts minus spaced, pooled (one-sided)	4.14	[3.60, 4.69]	<.001	3	Supported
H52^h	Spaced: fixed minus changing shape, pooled (two-sided; TOST ±0.3)	−0.25	[−0.35, −0.15]	<.001	3	Different (fixed lower); not equivalent
H53^h	Fixed-shape spaced nouns: $L(s^*) - L(s)$, pooled (one-sided)	5.84	[5.57, 6.11]	<.001	3	Supported
H54	At 1,000: noun-specific copy score of changing-shape bursts (TOST ±0.3)	0.08	[0.02, 0.14]	.015		Within ±0.3
H55^h	At 1,000: changing-shape bursts minus row-geometry prediction (one-sided)	3.67	[3.03, 4.32]	<.001	3	Supported
H56^h	At 1,000: queried-noun minus context-noun effect of changing vs fixed bursts (one-sided)	1.25	[0.94, 1.56]	<.001	3	Supported
M3B	85.6M models: unseen copy score minus 1 nat (one-sided)	5.96	[5.22, 6.69]	<.001		Passed
H57	85.6M models, at 1,000: changing-shape bursts minus fixed-shape bursts (one-sided)	4.18	[2.95, 5.41]	<.001		Supported
M3C	Changing-only corpora: unseen copy score minus 1 nat	5.16	[3.25, 7.08]	<.001		Passed
M3C	Fixed-only corpora: unseen copy score minus 1 nat	3.87	[2.84, 4.90]	<.001		Passed
H58	At 1,000: changing-shape bursts (changing-only) minus fixed-shape bursts (fixed-only) (one-sided)	5.51	[4.24, 6.78]	<.001	3	Supported
H59	At 1,000: changing-shape bursts, changing-only minus mixed corpora (two-sided; TOST ±0.3)	0.24	[−1.44, 1.92]	.712		Inconclusive
H59	At 1,000: fixed-shape bursts, fixed-only minus mixed corpora (two-sided; TOST ±0.3)	−1.00	[−1.20, −0.80]	<.001		Lower in fixed-only corpora

H52: TOST p = 0.14; 90% interval [−0.33, −0.17]. H54: TOST p <.001. H55, pooled noun-level regression with size and seed effects and seed-clustered errors: 3.82 [3.18, 4.46]; without no-shape nouns: 3.66 [3.05, 4.28]. By size: 6.36, 2.47, 2.19.

and no-shape nouns at 1,000 exposures and to unseen nouns with a typical row; the direction from 100-exposure bursts is added to fixed-shape bursts at 1,000 and subtracted from changing-shape bursts at 1,000, so the direction never comes from the nouns it is applied to. The control is a random vector of the same length (Table F4). *Activation patching*: for each fixed-shape burst noun F and the changing-shape burst noun R of the same index (1,000 exposures), the residual stream after a given block at the query noun's position, or at the last position, is copied from the item that names R in the query into the item that names F, and the reverse. After the first block, patching the query noun's position carried 5 to 58% of the difference between the two items (by size and direction) and patching the last position 47 to 96%; after the second block, the noun's position carried none and the last position all of it. *Attention*: first-layer attention from the last position to the context's shape word, mean over heads, was 0.069 [0.048, 0.089] higher for changing- than fixed-shape bursts at 1,000 exposures and tracked the copy score: across the 12 history-by-exposure cells within each model, Spearman correlations averaged 0.80 [0.76, 0.85] over seeds (0.91, 0.79 and 0.72 by size); pooled over conditions and models, 0.73, and in later layers −0.36 to 0.23. Full results: results/amendment11/MECHANISM.md in the supplementary material.

Natural text (exploratory, added after the results). To ask which regime natural text resembles, we counted the 44 nouns of the pretrained test in WikiText-103 (raw training split; 29,433 articles, 83.7 million words; scripts/natural_regime.py). Of a noun's mentions, a median 40% (colour-typical nouns) and 24% (colour-variable) were followed by another mention within 128 words, against a median 0.3% and 0.1% expected if mentions were spread evenly (under 1% for 42 of the 44 nouns; median ratios

Table F2: Experiment 3: copy score by corpus design, history and exposures (means over seeds by size, and seed-level mean with 95% interval, sizes averaged). Mixed: ten seeds; single-history designs: five seeds. [a]Over the three smaller sizes, as in the registered tests; 85.6M models were trained in the mixed design only.

Corpus	History	Exp.	3.4M	10.9M	25.6M	85.6M	Mean[a]	95% CI[a]
Mixed	Unseen	–	8.96	6.03	5.65	6.96	6.88	[6.31, 7.45]
	Fixed, spaced	100	1.10	1.53	2.10	1.29	1.58	[1.34, 1.82]
	Fixed, spaced	1,000	0.36	0.64	0.39	0.30	0.46	[0.27, 0.65]
	Fixed, bursts	100	6.17	3.76	4.36	5.12	4.76	[4.18, 5.35]
	Fixed, bursts	1,000	0.69	1.09	1.32	0.67	1.03	[0.83, 1.23]
	No shape, spaced	100	2.59	2.93	4.03	3.98	3.18	[2.68, 3.68]
	No shape, spaced	1,000	−0.02	0.92	1.67	0.91	0.86	[0.56, 1.15]
	No shape, bursts	100	3.16	4.66	5.27	5.81	4.36	[3.76, 4.96]
	No shape, bursts	1,000	0.06	1.55	1.54	1.16	1.05	[0.80, 1.30]
	Changing, spaced	100	1.63	2.12	2.17	2.18	1.97	[1.63, 2.32]
	Changing, spaced	1,000	0.28	0.63	0.79	0.36	0.57	[0.42, 0.71]
	Changing, bursts	100	9.03	4.53	4.77	5.55	6.11	[5.57, 6.66]
	Changing, bursts	1,000	7.05	3.53	3.56	4.85	4.71	[4.03, 5.40]
Changing only	Unseen	–	9.01	5.15	4.33	–	6.16	[4.25, 8.08]
	Changing, spaced	100	1.65	2.20	1.37	–	1.74	[0.96, 2.52]
	Changing, spaced	1,000	0.43	1.55	0.83	–	0.94	[0.44, 1.44]
	Changing, bursts	100	9.33	4.80	3.72	–	5.95	[4.09, 7.81]
	Changing, bursts	1,000	9.23	4.39	3.00	–	5.54	[4.12, 6.96]
Fixed only	Unseen	–	7.22	3.31	4.09	–	4.87	[3.84, 5.90]
	Fixed, spaced	100	0.18	1.05	1.26	–	0.83	[0.71, 0.95]
	Fixed, spaced	1,000	−0.16	0.11	0.08	–	0.01	[−0.10, 0.12]
	Fixed, bursts	100	1.68	1.55	2.09	–	1.77	[1.26, 2.29]
	Fixed, bursts	1,000	−0.23	0.31	0.00	–	0.03	[−0.13, 0.18]

Table F3: Experiment 3, mixed design, 1,000 exposures: copy score when the context sentence is about a noun of one history (rows) and the query names a different noun of the same or another history (columns); means over ten seeds, sizes averaged (24 items per pair and model).

Context \ query	Unseen	Fixed spaced	Fixed bursts	No shape spaced	No shape bursts	Changing spaced	Changing bursts
Unseen	6.87	0.67	0.97	0.99	1.16	0.51	2.98
Fixed, spaced	3.47	0.54	0.89	0.73	0.90	0.48	1.90
Fixed, bursts	5.23	0.71	1.13	0.94	1.16	0.60	3.02
No shape, spaced	4.12	0.40	0.85	0.83	1.00	0.41	1.82
No shape, bursts	4.84	0.61	0.94	0.87	0.87	0.51	2.71
Changing, spaced	4.08	0.47	0.91	0.93	1.08	0.56	2.16
Changing, bursts	7.09	0.81	1.28	1.32	1.44	0.80	4.73

Table F4: Experiment 3, steering (exploratory): copy score with the noun's own row (baseline; typical row for unseen nouns) and change when the changing-minus-fixed burst direction from other nouns, or a random vector of the same length, is added to the row (nats; ten seeds, sizes averaged). Nouns at 1,000 exposures. [a]Direction from 100-exposure bursts. [b]Direction from 100-exposure bursts subtracted.

Nouns	Baseline	Direction	95% CI	Random	95% CI
Fixed, spaced	0.46	1.51	[1.20, 1.82]	−0.02	[−0.11, 0.08]
Changing, spaced	0.57	1.06	[0.76, 1.36]	−0.01	[−0.06, 0.04]
No shape, spaced	0.86	2.04	[1.58, 2.49]	−0.01	[−0.16, 0.13]
No shape, bursts	1.05	3.06	[2.45, 3.67]	0.02	[−0.16, 0.19]
Unseen, typical embedding	0.79	1.99	[1.55, 2.44]	−0.02	[−0.24, 0.19]
Fixed, bursts[a]	1.03	0.99	[0.53, 1.44]	0.07	[−0.04, 0.19]
Changing, bursts[b]	4.71	−0.82	[−1.37, −0.28]	−0.30	[−0.59, −0.01]

133 and 221). We took a basic colour word immediately before the noun as a colour mention (2,341 for colour-typical nouns, 736 for colour-variable). Within an article, two colour mentions of the same noun named the same colour with median agreement 0.98 and 0.86 (nouns with at least three within-article pairs: 13 and 9); two mentions drawn from the whole corpus agreed with probability 0.46 and 0.37 (medians over nouns). Many colour–noun pairs are parts of names (*Black Sea*, *Blue Velvet*), which inflates agreement within articles, so these figures are rough. Per-noun values: results/natural_regime.csv in the supplementary material.

Tied embeddings (exploratory, added after the results). Ten 3.4M models were trained on the mixed-design corpora of Part A, with the same seeds and settings but with input and output embeddings tied (Apple M3 Ultra; the first model was checked after training and its input and output matrices were identical). With tied embeddings an unseen noun's row receives gradient as an output. Seed-level values, with the untied 3.4M models of Part A in brackets: unseen nouns copied 3.96 [2.43, 5.49] (8.96), and 0.02 [−0.18, 0.21] with a typical row (−0.19); at 1,000 exposures changing-shape bursts copied 1.90 [1.20, 2.60] more than fixed-shape bursts (6.36); changing-shape bursts copied 2.36 [1.47, 3.25] more than changing-shape spaced nouns, pooled (7.09); fixed- minus changing-shape spaced nouns, pooled, was −0.01 [−0.30, 0.28] (−0.23); and fixed-shape spaced nouns preferred their own shape to the context's by 4.61 [4.10, 5.11] (6.94). Familiar nouns of the other five histories kept more copying than in untied models (1.4 to 2.9 nats at 1,000 exposures, against 0 to 0.7), while changing-shape bursts kept less (4.29 against 7.05).